

Class 10 Science - Chapter 11

Electricity (Hinglish Notes)

PART A - CORE BASICS (MISS MAT KARNA)

Class 7-8 ki yeh cheezein pehle pakki karo, warna Ch11 samajh nahi aayega.

1. Charge aur Current ka idea (Class 8)

Electric charge = matter ka woh property jisse bijli ka asar hota.

Charge ka unit = COULOMB (C). Electron par -ve charge.

Electric current = charge ka flow (behaav). Unit = AMPERE (A).

Conductor = bijli pass kare (metal, copper). Insulator = na kare (rubber, plastic).

2. Circuit ka idea (Class 7-8)

Circuit = closed (band) loop jisme current behti.

Cell/Battery = current ka source (energy deti). Switch = on/off.

Open circuit = toota (current nahi); Closed = jud-a (current behti).

Current cell ke +ve se -ve terminal ki taraf behti (conventional).

3. Symbols (RATTA - circuit diagram ke liye)

Cell: lambi line (+), chhoti moti line (-). Battery: 2+ cells.

Resistor: zigzag ya rectangle box. Bulb: cross wala circle.

Ammeter (A): current naapta (series me). Voltmeter (V): voltage (parallel me).

4. Maths basics (numericals ke liye)

Formula me units sahi rakho (V volt, I ampere, R ohm).

Cross-multiply, fraction jodna ($1/R1 + 1/R2$), reciprocal lena aana chahiye.

Powers of 10: kilo = 1000, milli = 1/1000.

5. Energy aur Power (Class 8 idea)

Energy = kaam karne ki capacity (unit Joule J).

Power = energy use hone ki RATE (per second). Unit Watt (W). $1W = 1J/s$.

-- END OF CORE BASICS (RED) PAGE --

PART B - CHAPTER KA MAIN CONTENT

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[GREEN = board exam me baar-baar poocha jaata hai]

1. Electric Current aur Potential Difference [IMPORTANT]

Electric current (I) = charge flow ki rate. $I = Q / t$

Q = charge (coulomb), t = time (second). Unit: 1 ampere = 1 C/s.

1 electron charge = 1.6×10^{-19} C. (1 coulomb = $\sim 6.25 \times 10^{18}$ electrons.)

Potential difference (V) = do point ke beech charge ko move karne me kiya gaya kaam per unit charge. $V = W / Q$. Unit = VOLT (V). $1V = 1 J/C$.

Voltmeter parallel me, Ammeter series me lagta.

2. Ohm's Law [TOP EXAM AREA]

Ohm's Law: constant temperature par conductor me current (I) uske do siron ke beech voltage (V) ke SEEDHE proportional hoti.

$V = I R$ (R = resistance, constant).

Resistance (R) = current ke behaav ka virodh (opposition). Unit = OHM (symbol).

V vs I graph = SEEDHI line (straight line through origin) $\rightarrow$ slope = R.

3. Resistance kis par depend karta [VERY IMPORTANT]

R depend karta:

(1) Length (L) par: R seedha proportional (lamba taar $\rightarrow$ R jyada).

(2) Area (A) par: R ulta proportional (mota taar $\rightarrow$ R kam).

(3) Material par: resistivity (ρ) - har material ka apna.

(4) Temperature: badhne par R aam taur par badhta.

Formula: $R = \rho * L / A$ (ρ = resistivity, unit ohm-metre).

Resistivity: conductor (metal) bahut kam; insulator bahut zyada.

Alloys (nichrome, manganin) ki resistivity zyada + temperature se kam asar

$\rightarrow$ isliye heating element (heater/iron) me nichrome use hota.

4. Series Combination [TOP EXAM AREA]

Series = resistors ek ke baad ek (single path).

- Current (I) SAME har resistor me.

- Voltage baant-ta: $V = V_1 + V_2 + V_3$.

- Total resistance: $R_s = R_1 + R_2 + R_3$ (badhta jaata, sabse jyada).

Nuksaan: ek component fail $\rightarrow$ poora circuit band (jaise purani LED jhaalar).

5. Parallel Combination [TOP EXAM AREA]

Parallel = resistors alag-alag branch me (multiple path).

- Voltage (V) SAME har resistor par.

- Current baant-ta: $I = I_1 + I_2 + I_3$.
- Total resistance: $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$ (ghat-ta, sabse kam).

Faayda (ghar me parallel kyun):

- Har appliance ko poora 220V milta.
- Ek band ho to baaki chalte rehte.
- Har device ka apna switch + alag current.

6. Heating Effect of Current (Joule's Law) [TOP EXAM - NUMERICALS]

Current behne par resistor me HEAT paida hoti (Joule heating).

$H = I^2 R t$ (Joule's law of heating).

H = heat (J), I = current, R = resistance, t = time.

Heat current ke SQUARE par depend -> current double, heat 4 guna.

Uses: electric heater, iron, geyser, toaster, electric bulb (filament tungsten

- high melting point), FUSE.

FUSE = patli taar (kam melting point) jo zyada current par PIGHAL kar circuit tod deti -> appliance ko aag/damage se bachati. (Safety device.)

7. Electric Power [TOP EXAM - NUMERICALS]

Power P = energy per unit time.

$P = V I = I^2 R = V^2 / R$ (teeno form yaad rakho).

Unit: WATT (W). 1 W = 1 volt x 1 ampere. 1 kW = 1000 W.

Commercial unit of energy = kilowatt-hour (kWh) = "1 UNIT" bijli ka.

1 kWh = 1000 W x 3600 s = 3.6×10^6 J.

Bill: Energy (kWh) = Power(kW) x time(hours). Cost = units x rate.

8. Quick Revision - Formula Sheet [RATTA]

- $I = Q/t$; $V = W/Q$; Ohm: $V = IR$.
- $R = \rho L / A$ (lamba->R jyada, mota->R kam).
- Series: $R_s = R_1 + R_2 + \dots$ (current same); Parallel: $1/R_p = 1/R_1 + \dots$ (V same).
- Heat: $H = I^2 R t$. Power: $P = VI = I^2 R = V^2/R$.
- 1 unit = 1 kWh = 3.6×10^6 J. Fuse = safety (kam m.p. taar).

PART C - SOLVED EXAMPLES (HARDEST -> EASIEST)

Example 1 (HARDEST) - Series + Parallel mix + power + bill

Q: 6 ohm aur 3 ohm parallel me jude, phir 220V supply se. (a) Total R (b) Total current (c) Power (d) 5 ghante me kitne unit + Rs.8/unit par cost?

Solution:

(a) Parallel: $1/R_p = 1/6 + 1/3 = 1/6 + 2/6 = 3/6 = 1/2 \rightarrow R_p = 2 \text{ ohm}$.

(b) $I = V/R = 220/2 = 110 \text{ A}$.

(c) $P = VI = 220 \times 110 = 24200 \text{ W} = 24.2 \text{ kW}$.

(d) Energy = $24.2 \text{ kW} \times 5 \text{ h} = 121 \text{ kWh (units)}$. Cost = $121 \times 8 = \text{Rs. } 968$.

Example 2 - Heating effect numerical

Q: 5 ohm resistor me 2 A current 3 minute behti. Kitni heat paida hogi?

Solution:

- $H = I^2 R t$. $t = 3 \text{ min} = 180 \text{ s}$.

- $H = (2)^2 \times 5 \times 180 = 4 \times 5 \times 180 = 3600 \text{ J}$.

Example 3 - Resistance of wire (rho L/A)

Q: Ek taar ko kheech kar dugna (double) lamba kar diya (volume same). Naya resistance purane ka kitna guna?

Solution:

- Length double $\rightarrow L' = 2L$. Volume same $\rightarrow$ area aadha $\rightarrow A' = A/2$.

- $R = \rho L/A \rightarrow R' = \rho (2L)/(A/2) = 4 \rho L/A = 4R$.

- Naya resistance = 4 GUNA (kyunki R length² ke proportional jab volume same).

Example 4 - Ohm's law basic numerical

Q: 12 V battery se 4 ohm resistor juda. Current nikaalo.

Solution:

- $V = IR \rightarrow I = V/R = 12/4 = 3 \text{ A}$.

Example 5 (EASIEST) - Power formula

Q: 220 V par 2 A leta appliance ka power?

Solution:

- $P = VI = 220 \times 2 = 440 \text{ W}$.

PART D - SELF TEST (EASY -> HARD)

EASY (1 mark)

- Q1. Electric current ka formula + unit.
- Q2. Ohm's law likho.
- Q3. Fuse ka kaam kya hai?
- Q4. 1 unit (kWh) kitne joule ke barabar?

MEDIUM (2-3 marks)

- Q5. Resistance kis-kis par depend karta (formula sahit).
- Q6. Series aur parallel ka total R formula + ek-ek antar.
- Q7. Ghar me parallel connection kyun use hota (2 reason).
- Q8. Power ke teeno formula likho + unit.

HARD (3-5 marks, board favourite)

- Q9. 6 ohm + 3 ohm parallel, 220V: total R, current, power, 5h ka cost (Rs.8).
- Q10. Heating effect: 5 ohm, 2A, 3 min -> heat. Joule's law samjhao.
- Q11. Taar dugna lamba (volume same) -> resistance kitna guna?
- Q12. Heating element me nichrome (alloy) kyun use hota - 2 reason.

ANSWER HINTS (PART D)

- A1. $I = Q/t$; unit ampere (A).
- A2. $V = IR$ (constant temp par I, V ke proportional).
- A3. Zyada current par pighal kar circuit tod deti (safety).
- A4. $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$.
- A5. Length (seedha), area (ulta), material (ρ), temp; $R = \rho L/A$.
- A6. Series $R_s = R_1 + R_2$ (current same); Parallel $1/R_p = 1/R_1 + 1/R_2$ (V same).
- A7. Har appliance poora 220V; ek band ho to baaki chale.
- A8. $P = VI = I^2 R = V^2/R$; unit watt.
- A9. $R_p = 2 \text{ ohm}$, $I = 110 \text{ A}$, $P = 24.2 \text{ kW}$, units=121, cost=Rs.968.
- A10. $H = I^2 R t = (4)(5)(180) = 3600 \text{ J}$; heat current^2 par depend.
- A11. 4 guna (L double, area aadha).
- A12. High resistivity + temp se kam asar (jyada garmi deta, oxidise nahi).

"Padhai tabhi pakki jab khud likh ke test do." - All the best, Krishna!

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